PHS 2000B Problem Set 8: Interrupted Time Series and Difference-in-Differences

Due April 16, 2026 at 11:59pm

Part 1: True, false, or uncertain

Are the following statements true, false, or uncertain. Briefly explain your answer (note: if you answer False, it is not sufficient to simply state the opposite of the claim).

Question 1

Claim: You are leveraging an ITS approach in order to evaluate the effect of a national policy change to child tax credits on childhood malnutrition. You know that two years after the tax credit policy, there were also national expansions school feeding programs. By adjusting your model for this second policy change, you are certain to you can still get unbiased estimates. (2 points)

Solution

False. An essential assumption for causal inference when using ITS is that there are no timevarying confounders.

Question 2

Claim: If we have longer time series data, then we will always be more likely to get unbiased estimates with an interrupted time series approach. (2 points)

Solution

False. The length of the time series is unrelated to the identifying assumption of an ITS analysis, which is that the timing of the intervention is as good as random. A longer time series might, in fact, make it more difficult for us to model time trend as well due to various other factors that are changing over time.

Question 3

Claim: In CITS, the control group does not need to have the same pre-trends as the treated group. (2 points)

Solution

True. While both the control and treated groups having similar trends in the pre-period gives us more confidence that the control group is appropriate, the only requirement is that changes in trends in the post-period would have been the same between the two in the absence of treatment and the only reason they would be different is due to treatment

Part 2: Mix and match scenarios with event study plots

Match the each scenario with the event study plot that best illustrates the estimates of the effects estimated and explain why. Comment on the effects estimated and your confidence in the parallel trends assumptions (2 sentences, 3 points each).

Question 4

Using data from hospitals in your local health system, you evaluate the evaluate the effect of a physician training program implemented in a subset of the hospitals on the rate of hospital-acquired infections (HAIs). You find that the program was effective in reducing HAIs while the trainings were ongoing, but there seems to be no lasting effect after.

Solution Plot B best matches this scenario. In this plot, we see that there is a statistically significant reduction in the outcome during the first three periods of the intervention being evaluated. However, the remaining estimated period-specific ATTs in the treated period hover around 0 and are not statistically significant, indicating the effect fades out. Based on the pre-period ATTs hovering around the null line and not being statistically different from zero, it seems the pre-trends are parallel, which reassures the validity of the parallel trends assumption.

Question 5

There is a hypothetical policy implemented in some states where sexually transmitted infection (STI) testing is made free everyone in the state. You know that every state that implemented this policy did so because they experienced unexpected high counts of STI transmissions in the periods just prior policy implementation. If one were to observe both treated and untreated counterfactuals for those states that passed this policy, they would able to observe that the policy had no effect.

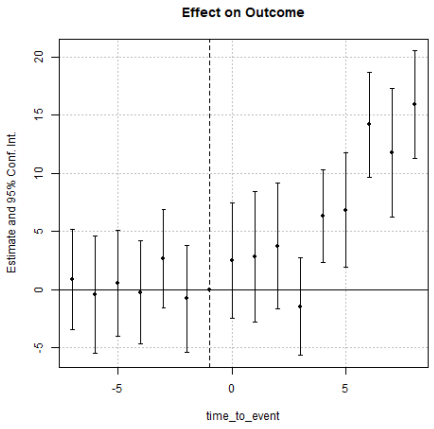
Solution Plot C best matches this scenario. In this plot, we see evidence of violations to the parallel trends assumption in the pre-period. While the ATT estimates in the post-period indicate that the ploicy is effective in reducing STI transmission, this only further highlights that violations to the parallel-trends assumption induces bias into our estimates.

Question 6

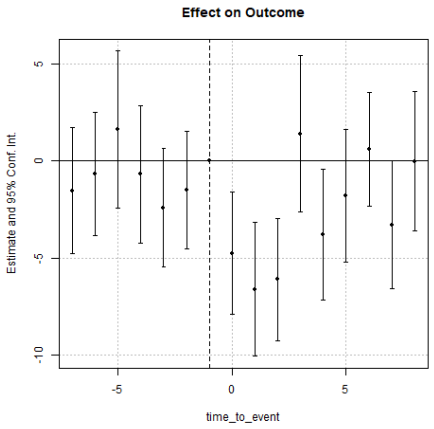
You are evaluating the effect of rural towns establishing mobile vaccination clinics to better serve hard-to-reach communities. You use monthly data of total number of MMR vaccines administered to children aged 5 years and younger. You find that these mobile clinics were eventually effective, but it took a few months for people living in these communities to find out about them and, subsequently, utilize their services

Solution Plot A best matches this scenario. This plot illustrates that there is no significant effect initially in post-intervention period, but the positive impact on the outcome seems to be increasing as time under treatment goes on.

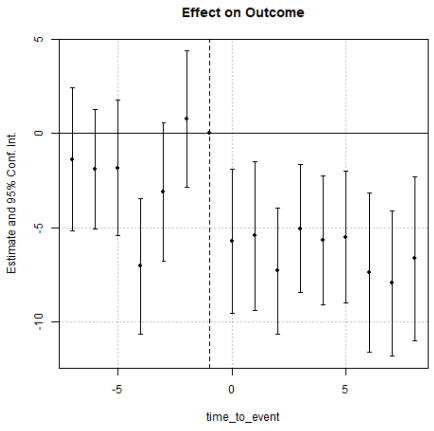
A.


Effect on Outcome
Estimate and 95% Conf. Int.
time_to_event

B.


Effect on Outcome
Estimate and 95% Conf. Int.
time_to_event

C.


Effect on Outcome
Estimate and 95% Conf. Int.
time_to_event

Part 3: Conducting an Difference-in-Difference analysis - School feeding program

We will now use R to conduct a difference-in-difference analysis. For these questions, you will be asked to run analyses and create plots.

Many countries are interested the impact of providing students with free meals in schools on their academic performance. Firstly, malnutrition impacts attention in the classroom and increases the likelihood of being absence due to sickness via weakened immune systems. Conversely, a reliable source of food incentives families to ensure that their children make it to school. Use the provided dataset to evaluate a hypothetical school feeding program. In the data, there are 50 villages, with 10 schools in each village. Half of the villages instituted a school feeding program in every school in the village during the eighth time period. Data on the total number absences in each school is recorded on a monthly basis, over the span of 16 months.

Variable Name	Variable Type	Explanation
treat	Dichotomous	indicates whether the village ever implements the program
post	Dichotomous	Whether the observation is from interval 8 or later
post_treat	Dichotomous	treat X post
village	Categorical	Village ID number
interval	Continuous	Time period, measured in months
time_to_event	Continuous	Time period, measured in months
interval	Continuous	Time period, measured in months
time_to_event	Continuous	Number of months from interval of program start; coded as 0 for non-treated villages
absences	Continuous	Number of student absences per school-month

Question 7

Plot the trends in school absences over time by treatment group Your graph should include the following elements:

- Labels for your x- and y-axis
- A vertical line in the color of your choice, noting when the intervention started
- A line connecting average school absences per month for each treatment group

Describe the overall trends in the pre and post period and how they compare between the two groups. (3 points)

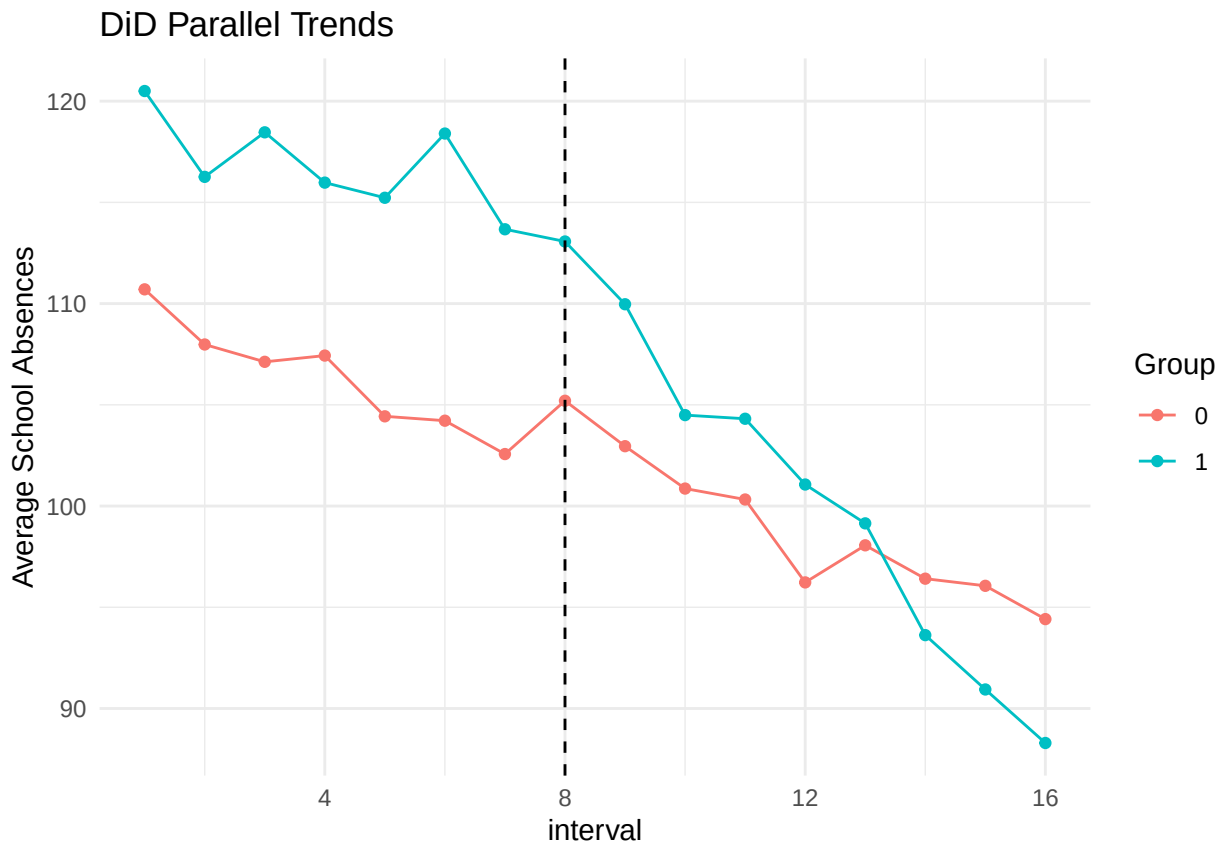
```
# Please install these packages if you don't have them already
library(ggplot2)
library(fixest)
library(dplyr)

# Loading data set
did_data <- read.csv("PHS2000B_pset8_data.csv")         # Change here if you have renamed the file

# 1. Calculate average outcome by treatment group and month
plot_data <- did_data %>%
  group_by(interval, treat) %>%
  summarize(absences_total = mean(absences_total, na.rm = TRUE))

# 2. Create the parallel trends plot
ggplot(plot_data, aes(x = interval, y = absences_total, color = factor(treat))) +
  geom_line() +
```

```
  geom_point() +
  geom_vline(xintercept = 8, linetype = "dashed") + # Vertical line at treatment start
  labs(title = "DiD Parallel Trends", y = "Average School Absences", color = "Group") +
  theme_minimal()
```



Question 8

Estimate the average effect of the school feeding program using a TWFE model. Interpret the estimate of interest (2 point)

```
ATT<-feols(absences_total ~ post_treat | school + interval, cluster = ~village, data = did_data)
summary(ATT)
```

```
## OLS estimation, Dep. Var.: absences_total
## Observations: 8,000
## Fixed-effects: school: 500,  interval: 16
## Standard-errors: Clustered (village)
##            Estimate Std. Error  t value   Pr(>|t|)
## post_treat -8.97848       1.02 -8.80246 1.1636e-11 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
## RMSE: 19.6     Adj. R2: 0.243625
##               Within R2: 0.012766
```

Solution

The school feeding program reduced the average monthly count of student absences by 9 absences duing

the post period in schools that implemented the program. This effect is statistically significant at an alpha-threshold of 0.05.

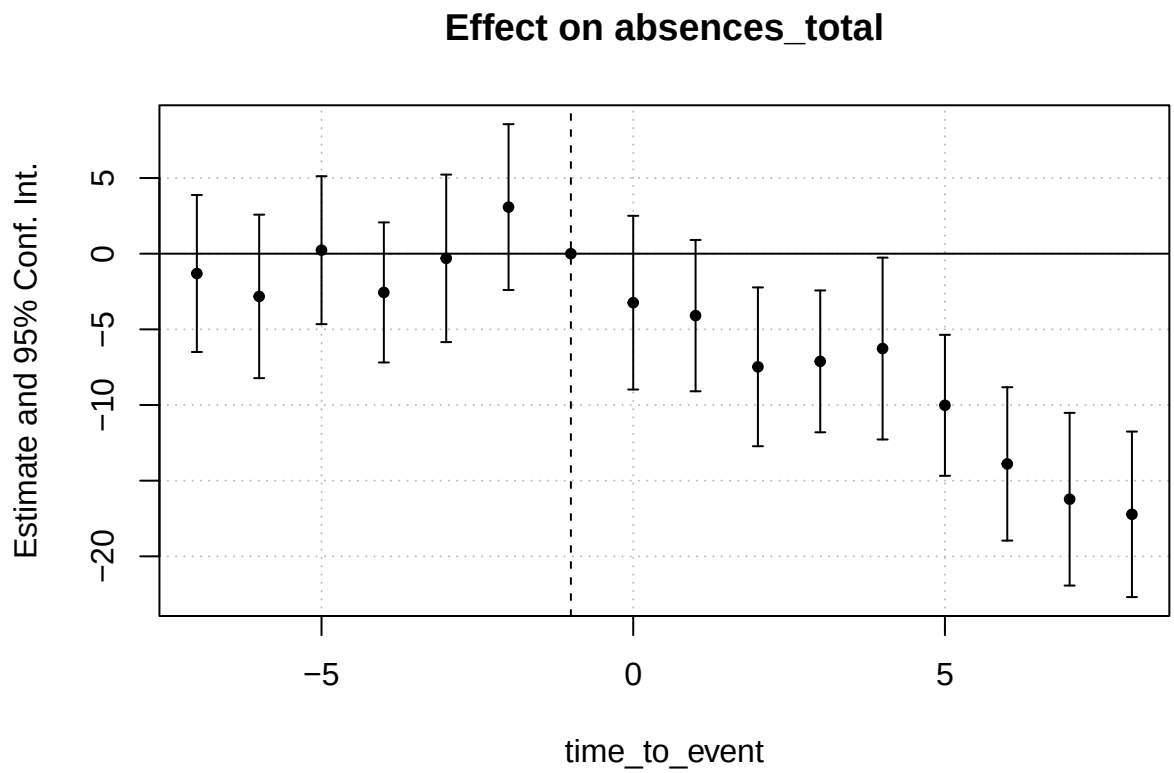
Question 9

Now, using a TWFE model, estimate a model allowing for period-specific ATTs and plot the results. What additional information does this model provide as opposed to the previous model estimated? (2 points)

Solution

Based on the plot we have just created, we can see that the effect of program gradually grew in magnitude, with the first statistically significant period-specific ATT being in the third period of program implementation.

```
eventstudy <- feols(absences_total ~ i(time_to_event, treat, ref = -1) | school + interval, cluster =~v
iplot(eventstudy)
```



Question 10

Using the event study graph you just created, comment on the implications of the estimates in the pre-period on your confidence of the parallel trends assumption in this study? (2 points)

Solution

Because all the estimated ATTs in the pre-period hover around the null line, we can say that the trends in outcomes for those schools that implemented this program and those that did not are parallel in the pre-period. However, this does not guarantee that these trends would have continued to be parallel in the post-period in the absence of the program.

Question 11

After running the analysis, you come to find out that parents are aware of which villages have a school feeding program and are able enroll to freely choose where to send their children. Does affect our confidence that our causal estimates are unbiased? Explain why or why not. (2 sentences, 2 points)

Solution

If this were the case, we would be worried that certain sub-groups would select to move their children from schools that do not provide meals to schools that due. Under the assumption that these children might have different absenteeism behavior, this would be considered a SUTVA violation, where the treatment status of one school can affect the outcomes of another school.

Question 12

Why would we want to cluster our standard errors at the village level rather than the school level? (1 point)

Solution

While clustering at the school-level would likely shrink our standard errors further because there are more schools than there are villages, we must cluster our standard errors at the level we believe treatment was assigned in order to have valid statistical inference. Here, we know that whether or not a school should adopt this program was determined at the village-level, thus we cluster at this level.